

Dear Editors,

We are pleased to submit our manuscript titled "**Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks**" for your consideration in **IEEE Access Journal**.

In this paper, we present a comprehensive investigation into the field of Electricity Theft Detection (ETD) for smart homes. Our research is driven by the ever-growing importance of securing and optimizing electricity consumption, especially in the context of smart grid technologies. To address the pressing need for more effective ETD systems, we have explored the utility of machine learning models in detecting abnormal consumption patterns indicative of electricity theft.

One of the distinguishing features of our study is the inclusion of both real-world and synthetic attack datasets. We leveraged real-world attack data to evaluate our models and test their performance in authentic scenarios. This inclusion allows us to draw valuable insights into the practical applicability of our models in real-world settings.

Our findings highlight the impressive performance of supervised learning models, particularly XGB and Random Forest, RF, when trained and tested on real attack datasets with average overall AUC scores of 98.74% and 98.69% for XGB; and 97.07% and 96.76% for RF in real and simulated attacks ETD respectively for all homes. This emphasizes the significance of incorporating real attack data into the training process, as it helps capture the nuances of real-world electricity theft.

In contrast, we also explored the potential of legacy unsupervised learning models (LUM), including MLP-AE, 1D-CONV-AE, and Isolation Forest and their performance on both real and synthetic datasets. Our proposed model Synthetic Discriminator Model (SYNDM) achieve higher detection rate than the legacy unsupervised model (LUM).

The conference version of this manuscript and the differences with the submitted manuscript are as follows:

1. Olufemi Abiodun Abraham, Hideya Ochiai, Md Delwar Hossain, Yuzo Taenaka, Youki Kadobayashi, "Electricity Theft Detection for Smart Homes with Knowledge-Based Synthetic Attack Data", in 2023 IEEE 19th International Conference on Factory Communication Systems (WFCS)

The conference version of this paper proposed ETD in three smart homes with different appliances configuration. We simulated 5 different attack scenarios that attacker could deploy to steal electricity in apartments, congested complex building or office rooms where power line cables are not easily traceable. We deployed 9 different machine learning algorithms to classify each of this scenario and recommend the best performing models.

We extend this manuscript by first preprocess our dataset for binary classification and then augmenting benign records from our original dataset for training the models to be able to classify between anomaly and abnormal instances.

2. We provide a detailed explanation of the proposed synthetic binary discriminator for ETD and the legacy unsupervised models architectures and compare results in different smart homes in the journal manuscript.

3. We thoroughly discuss the dataset processing and use real attack dataset from a building to evaluate our simulated attacks through systematic experimentation and its impact on detection accuracy.

4. We deeply elaborate on why the proposed SYNDM is effective for smart home consumption pattern attack detection. We also discussed the limitations.

5. In the journal manuscript, we consider our research contribution to the development of more effective ETD systems for smart homes. By addressing the limitations of our models and exploring these future research directions, we aim to enhance the security and reliability of electricity consumption monitoring systems.

In this article, we clearly describe how this work differs from our previous conference paper. The manuscript has not been published elsewhere and is not under consideration by another journal. I/We have approved the manuscript and agree with submission to IEEE Access Journal. We believe our study aligns with the scope and interests of IEEE Access and would be a valuable addition to the journal.

Sincerely,

Olufemi Abiodun Abraham